

大学物理甲 II 习题参考解答 (第九章 真空中的静电场 II)

第九章

9-8 将半球面分割成许多极窄的三维圆环, 其带电量及圆环在球心所产生的场强为:

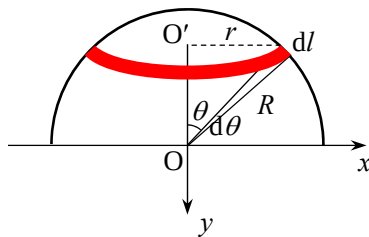
$$dq = \sigma dS = \sigma \cdot 2\pi r dl; \quad dl = R d\theta; \quad r = R \sin \theta;$$

$$dq = \sigma \cdot 2\pi R^2 \sin \theta d\theta$$

该圆环在 O 点产生的场强

$$dE = \frac{dq}{4\pi\epsilon_0 R^2} \cos \theta = \frac{\sigma \sin \theta \cos \theta d\theta}{2\epsilon_0};$$

$$E = \frac{\sigma}{2\epsilon_0} \int_0^{\pi/2} \sin \theta \cos \theta d\theta = \frac{\sigma}{4\epsilon_0}$$



9-9 选取 o 点为坐标原点, y 轴竖直向上, x 轴水平向右:

半无限长直导线 A 在 O 点产生的场强 E_1 为:

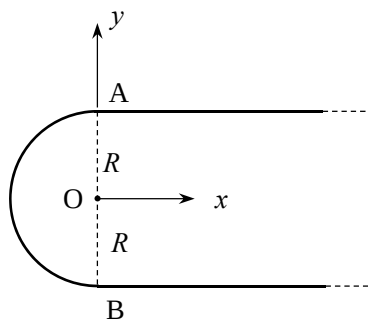
$$\vec{E}_1 = -\frac{\lambda}{4\pi\epsilon_0 R} \vec{i} - \frac{\lambda}{4\pi\epsilon_0 R} \vec{j};$$

半无限长直导线 B 在 O 点产生的场强 E_2 为:

$$\vec{E}_2 = -\frac{\lambda}{4\pi\epsilon_0 R} \vec{i} + \frac{\lambda}{4\pi\epsilon_0 R} \vec{j};$$

半圆弧在 O 点产生的场强 $E_{\text{弧}}$ 为

$$\vec{E}_{\text{弧}} = \frac{\lambda}{2\pi\epsilon_0 R} \vec{i}$$



根据场强叠加原理, O 点的场强为:

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_{\text{弧}} = 0$$

9.24 (1) O 点电势是各点电荷单独存在时在该处产生电势的代数和, 得

$$U_o = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r} + \frac{q_2}{r} + \frac{q_3}{r} + \frac{q_4}{r} \right) = 9 \times 10^9 \frac{4 \times 2.0 \times 10^{-9}}{5.0 \times 10^{-2}} = 1.44 \times 10^3 (\text{V})$$

(2) 电场力做功为

$$A = q_0 (U_\infty - U_o) = -q_0 U_o = -1.44 \times 10^{-6} (\text{J})$$

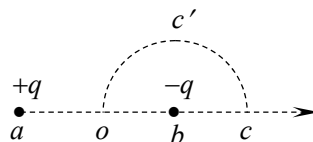
(3) 电势能的改变为

$$\Delta W = W_o - W_\infty = W_o = 1.44 \times 10^{-6} (\text{J})$$

$$9.25 \quad (1) \quad U_o = \frac{q}{4\pi\epsilon_0 l} - \frac{q}{4\pi\epsilon_0 l} = 0 \quad U_c = \frac{q}{4\pi\epsilon_0 3l} - \frac{q}{4\pi\epsilon_0 l} = -\frac{q}{6\pi\epsilon_0 l}$$

$$A_{oc} = q_0 (U_o - U_c) = \frac{qq_0}{6\pi\epsilon_0 l}$$

$$(2) \quad A_{c\infty} = -q_0 (U_c - U_\infty) = \frac{qq_0}{6\pi\epsilon_0 l}$$

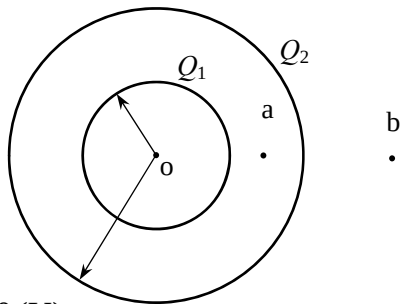


9.28 解法一：由高斯定理，可得电场分布为：

$$E_1 = 0 \quad (r < R_1)$$

$$E_2 = \frac{Q_1}{4\pi\epsilon_0 r^2} \quad (R_1 < r < R_2)$$

$$E_3 = \frac{Q_1 + Q_2}{4\pi\epsilon_0 r^2} \quad (r > R_2)$$



$$\text{a 点电势为: } U_1 = \int_{0.2}^{0.3} \vec{E}_2 \cdot d\vec{r} + \int_{0.3}^{\infty} \vec{E}_3 \cdot d\vec{r} = 900 \text{ (V)}$$

$$\text{b 点电势为: } U_2 = \int_{0.5}^{\infty} \vec{E}_3 \cdot d\vec{r} = 450 \text{ (V)}$$

$$\text{解法二: } U_{\text{小a}} = \frac{Q_1}{4\pi\epsilon_0 r_a} \quad U_{\text{大a}} = \frac{Q_2}{4\pi\epsilon_0 R_2}$$

$$\text{由电势叠加原理: } U_1 = \frac{Q_1}{4\pi\epsilon_0 \times 0.2} + \frac{Q_2}{4\pi\epsilon_0 \times 0.3} = 900 \text{ (V);}$$

$$U_{\text{小b}} = \frac{Q_1}{4\pi\epsilon_0 r_b} \quad U_{\text{大b}} = \frac{Q_2}{4\pi\epsilon_0 r_b}$$

$$\text{由电势叠加原理: } U_2 = \frac{Q_1}{4\pi\epsilon_0 \times 0.5} + \frac{Q_2}{4\pi\epsilon_0 \times 0.5} = 450 \text{ (V)}$$

$$9.29 \quad \text{点电荷的电势: } dU = \frac{dq}{4\pi\epsilon_0 x} = \frac{\lambda dx}{4\pi\epsilon_0 x}$$

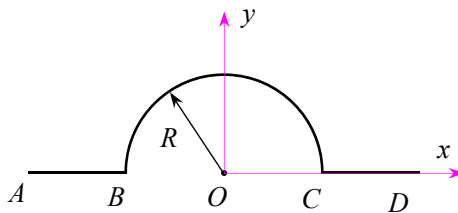
电势叠加原理：

$$U_{AB} = U_{CD} = \int_R^{2R} \frac{\lambda dx}{4\pi\epsilon_0 x} = \frac{\lambda}{4\pi\epsilon_0} \ln 2;$$

电势叠加原理：

$$U_{BC} = \int_0^Q \frac{dq}{4\pi\epsilon_0 R} = \frac{1}{4\pi\epsilon_0 R} \int_0^Q dq = \frac{\lambda \pi R}{4\pi\epsilon_0 R} = \frac{\lambda}{4\epsilon_0}$$

$$\text{电势叠加原理: } U_0 = U_{AB} + U_{BC} + U_{CD} = \frac{\lambda}{4\pi\epsilon_0} (2 \ln 2 + \pi)$$



$$9.30 \quad U_+ = \frac{q}{4\pi\epsilon_0 \sqrt{(x-l/2)^2 + R^2}} \quad U_- = \frac{-q}{4\pi\epsilon_0 \sqrt{(x+l/2)^2 + R^2}}$$

$$U = U_+ + U_- = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{(x-l/2)^2 + R^2}} - \frac{1}{\sqrt{(x+l/2)^2 + R^2}} \right)$$

当 $l \ll R$ 时，用泰勒级数在 $l=0$ 处展开，略去 l 的二次项，得： $U = \frac{qlx}{4\pi\epsilon_0 (x^2 + R^2)^{3/2}}$

$$9.31 \quad (1) \quad dU = \frac{dq}{4\pi\epsilon_0 r} = \frac{(Q/2l)dx}{4\pi\epsilon_0 \sqrt{x^2 + y^2}}$$

$$U = \int_{-l}^l \frac{(Q/2l)dx}{4\pi\epsilon_0 \sqrt{x^2 + y^2}} = \frac{Q}{4\pi\epsilon_0 l} \ln \frac{\sqrt{l^2 + y^2} + l}{y}$$

由对称性分析, 场强沿 y 轴:

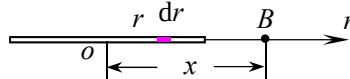
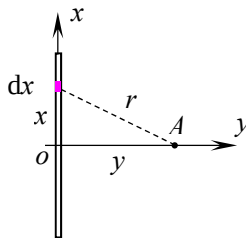
$$E = E_y = -\frac{dU}{dy} = \frac{Q}{4\pi\epsilon_0 y \sqrt{l^2 + y^2}}$$

$$(2) \quad dU = \frac{dq}{4\pi\epsilon_0 (x-r)} = \frac{Qdr}{8\pi\epsilon_0 l(x-r)}$$

$$U = \int_{-l}^l \frac{Qdr}{8\pi\epsilon_0 l(x-r)} = \frac{Q}{8\pi\epsilon_0 l} \ln \frac{x+l}{x-l}$$

由电荷分布, 可知电场沿 x 方向, 故

$$E = E_x = -\frac{dU}{dx} = \frac{Q}{4\pi\epsilon_0 (x^2 - l^2)}$$



$$9.32 \text{ 解法一: 锥侧长为变量: } dS = 2\pi r dl = 2\pi l \sin \theta dl$$

$$dq = \sigma 2\pi l \sin \theta dl \quad dU = \frac{2\pi l \sigma \sin \theta dl}{4\pi\epsilon_0 l}$$

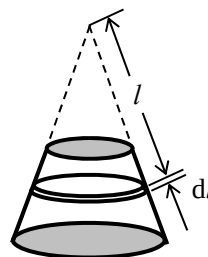
$$U = \int_{R_1/\sin \theta}^{R_2/\sin \theta} \frac{2\pi l \sigma \sin \theta dl}{4\pi\epsilon_0 l} = \frac{\sigma}{2\epsilon_0} (R_2 - R_1)$$

$$\text{解法二: 锥高为变量: } dS = 2\pi r dl = 2\pi x \tan \theta \frac{dx}{\cos \theta}$$

$$U = \int_{R_1/\tan \theta}^{R_2/\tan \theta} \frac{2\pi x \sigma \tan \theta \frac{dx}{\cos \theta}}{4\pi\epsilon_0 \frac{x}{\cos \theta}} = \frac{\sigma}{2\epsilon_0} (R_2 - R_1)$$

$$\text{解法三: 半径为变量: } dS = 2\pi r dl = 2\pi r \frac{dr}{\sin \theta}$$

$$U = \int_{R_1}^{R_2} \frac{2\pi r \sigma \frac{dr}{\sin \theta}}{4\pi\epsilon_0 \frac{r}{\sin \theta}} = \frac{\sigma}{2\epsilon_0} (R_2 - R_1)$$

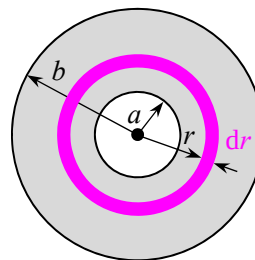


$$9.33 \quad \sigma = \frac{Q}{\pi(b^2 - a^2)};$$

$$dq = \sigma \cdot 2\pi r dr$$

$$dU = \frac{dq}{4\pi\epsilon_0 r} = \frac{\sigma}{2\epsilon_0} dr$$

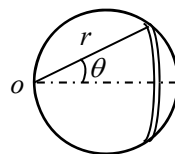
$$U = \int_a^b \frac{\sigma}{2\epsilon_0} dr = \frac{\sigma}{2\epsilon_0} (b - a) = \frac{Q}{2\pi\epsilon_0 (a + b)} = 9 \times 10^3 \text{ (V)}$$



9.34 解法一: $dq = \sigma \cdot 2\theta dr$ $dU = \frac{dq}{4\pi\epsilon_0 r} = \frac{\sigma\theta}{2\pi\epsilon_0} dr$

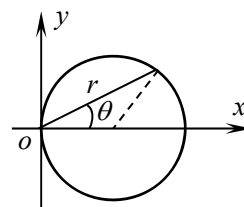
$r = 2R \cos \theta$ $dr = -2R \sin \theta d\theta$

$U = -\left(\frac{\sigma R}{\pi\epsilon_0}\right) \int_{\pi/2}^0 \theta \sin \theta d\theta = \frac{\sigma R}{\pi\epsilon_0}$



解法二: $dq = \sigma \cdot r dr d\theta$ $dU = \frac{dq}{4\pi\epsilon_0 r} = \frac{\sigma}{4\pi\epsilon_0} dr d\theta$

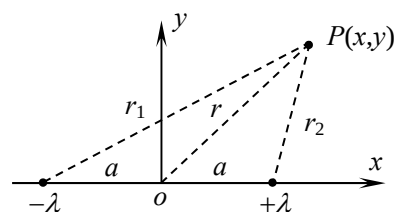
$U = \frac{\sigma}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} \left(\int_0^{2R \cos \theta} dr \right) d\theta = \frac{\sigma}{4\pi\epsilon_0} \int_{-\pi/2}^{\pi/2} 2R \cos \theta d\theta = \frac{\sigma R}{\pi\epsilon_0}$



9.35 $U_+ = \int_{r_2}^a \vec{E}_+ \cdot d\vec{r} = \int_{r_2}^a \frac{\lambda}{2\pi\epsilon_0 r} dr = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{a}{r_2}$

$U_- = \int_{r_1}^a \vec{E}_- \cdot d\vec{r} = \int_{r_1}^a \frac{-\lambda}{2\pi\epsilon_0 r} dr = \frac{-\lambda}{2\pi\epsilon_0} \ln \frac{a}{r_1}$

$U_P = U_+ + U_- = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_1}{r_2} = \frac{\lambda}{4\pi\epsilon_0} \ln \frac{(x+a)^2 + y^2}{(x-a)^2 + y^2}$



9.36 $U_P = \frac{q_1}{4\pi\epsilon_0 r_1} - \frac{q_2}{4\pi\epsilon_0 r_2} = 0$ $\frac{r_1}{r_2} = \frac{q_1}{q_2}$

设 A 点为直角坐标系的原点, P 点的坐标为 (x, y, z):

$\frac{\sqrt{x^2 + y^2 + z^2}}{\sqrt{(x-d)^2 + y^2 + z^2}} = \frac{q_1}{q_2}$

$x^2 - \frac{2q_1^2 d}{q_1^2 - q_2^2} x + y^2 + z^2 = -\frac{q_1^2 d^2}{q_1^2 - q_2^2}$

$\left(x - \frac{q_1^2 d}{q_1^2 - q_2^2}\right)^2 + y^2 + z^2 = \frac{q_1^2 q_2^2 d^2}{(q_1^2 - q_2^2)^2}$

